Qian 2020 ovarian cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

4/4 tumours (920–3345 malignant cells) intermix on one continuous manifold at Harmony θ=10 (per-neighbourhood iLISI 1.87 of a 2.89 ceiling); radius-graph: 0 detached islands.

Gate 2 — module geography → GO (non-branching)

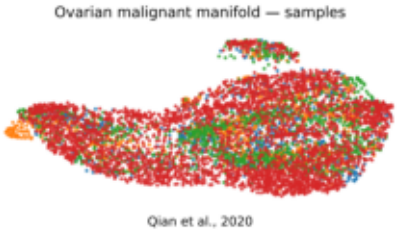
Gast/pSom co-vary (r=+0.36) and decline along pseudotime (ρ=−0.48/−0.54); Crest anti-correlated (r=−0.28/−0.43), rising to a terminal readout (ρ=+0.48; crest-high decile pt=0.68). Single (Gast/pSom)→Crest developmental axis.

Gate 3 — cross-repression → NA

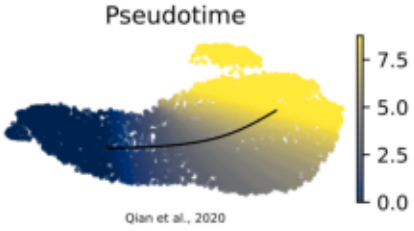
Single monotone axis — a PPT tree's arms co-vary in Gast/pSom (r=+0.68), so there is no bifurcation and no Gast/pSom hub; no branch pair exists to test for cross-repression.

STEP FIGURES

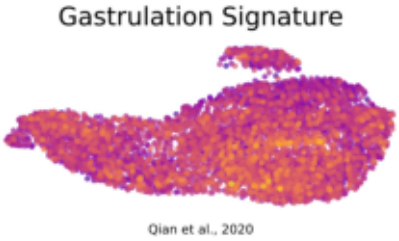
Manifold — samples



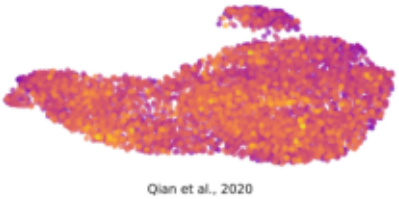
Pseudotime



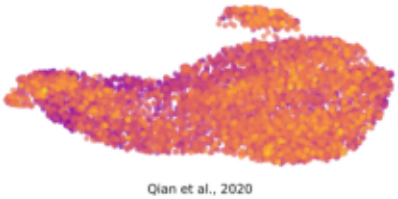
Gastrulation



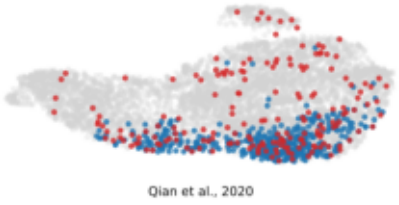
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Ovarian cancer (Qian et al., 2020), 10x — 4 tumours / 16,951 cells (master scope); malignant only: 6,490 load across all 4 samples (no thin tail to threshold; 0 islands removed). Whole malignant manifold (no lobe scoping). Non-branching Linear-pSom/Crest.

Verdict.
Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Qian's ovarian malignant manifold is a single monotone Gast/pSom → terminal-Crest gradient (Linear-pSom/Crest), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The developmental axis is orthogonal to the proliferation split (holds among non-cycling cells alone), so the whole manifold is kept (Lee/Bischoff no-scoping precedent). EMT-TFs PRRX1 (early/root) → PRRX2 (late); all TFs decline from the mesenchymal origin.

Parameter choices

Min cells/sample	≥100 (keeps 4/4; no thin tail)	**Integration**	Harmony θ=10; PCA 20; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=15, λ=0.3, μ=100
Root node	tip = Gast/pSom-high, Crest-low	**Compartments**	non-cycling 4,971 / cycling 1,519 (axis orthogonal to split)
EMT-TFs	PRRX1 (early/root) → PRRX2 (late); SNAI1/FOXC2 near detection floor		